Question:

Write a program to find the Valid Parentheses

Given a string containing just the characters '(', ')', '{', '}', '[' and ']', determine if the input string is valid. An input string is valid if:

1. Open brackets must be closed by the same type of brackets.
2. Open brackets must be closed in the correct order.

Note that an empty string is also considered valid.

Algorithm:

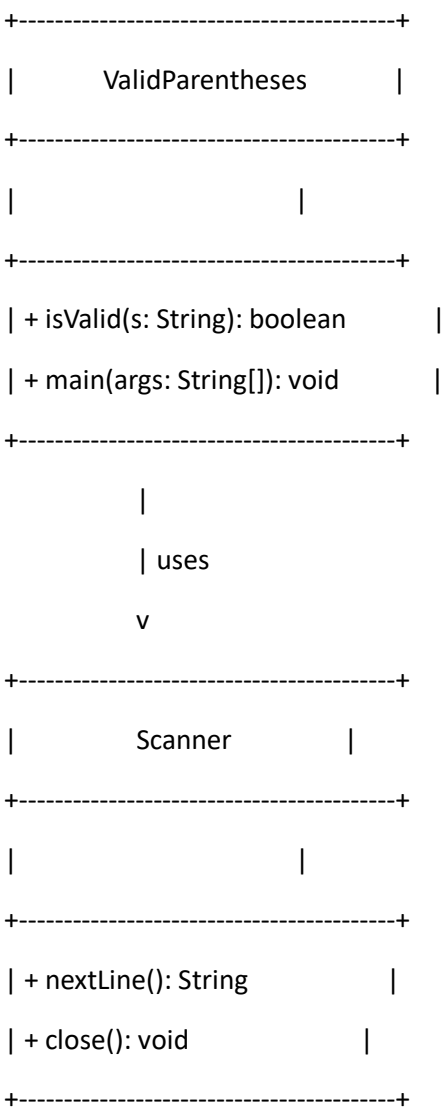
1. Start.
2. Read the bracket string from the user.
3. Create a character stack of size equal to
   the length of the string.
4. Initialize top = -1.
5. Traverse each character in the string.
6. If the character is an opening bracket
   '(', '{', or '[':
      Push it onto the stack.
7. Otherwise:
      a) If the stack is empty, return false.
      b) Pop the top opening bracket.
      c) Check whether the opening and closing
         brackets match.
      d) If they do not match, return false.
8. After traversing all characters:
      If top == -1, return true.
      Otherwise, return false.
9. In the main() method:
      If isValid() returns true:
         Display "Valid".
      Otherwise:

Display "Not valid".

10. Close the Scanner.

11. Stop.

UML Class Diagram:



Code:

```
package javacore;

import java.util.Scanner;

public class ValidParentheses {
```

```
    public static boolean isValid(String s) {
        char[] stack = new char[s.length()];
        int top = -1;

        // Check each character in the string
        for (int i = 0; i < s.length(); i++) {
            char ch = s.charAt(i);

            // Push opening brackets into the stack
            if (ch == '(' || ch == '{' || ch == '[') {
                stack[++top] = ch;
            }

            // Check closing brackets
            else {
                // No opening bracket is available
                if (top == -1) {
                    return false;
                }

                char open = stack[top--];

                // Check whether brackets match
                if ((ch == ')' && open != '(') ||
                    (ch == '}' && open != '{') ||
                    (ch == ']' && open != '[')) {
                    return false;
                }
            }
        }
    // Valid only if no opening brackets remain
```

```
        return top == -1;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter brackets: ");
        String s = sc.nextLine();

        if (isValid(s)) {
            System.out.println("Valid");
        } else {
            System.out.println("Not valid");
        }

        sc.close();
    }
}
```

output

```
Enter brackets: ()
Valid
```